



Habib University

EE-426/CE-442 Cryptography and Network Security - Fall 2024

Instructor: Farhan Khan

Time = 75 minutes

Major Exam 02

Max Points: 100

Instructions:

- i. **Smart watches and similar electronics are strictly NOT allowed. You are allowed to use dcode.fr website on your laptop for modular exponentiation operations.**
- ii. **Answer sheets should contain all steps, working, explanations, and assumptions.**
- iii. Double-sided A4-sized cheat sheet is allowed.
- iv. Attempt the exam on clean papers with black/blue ink.
- v. Write your name and HU ID on answer sheet.
- vi. Turn in your question paper along with your answer sheet.
- vii. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment: This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply the concepts from number theory and algebraic structures in understanding the design of cryptographic algorithms.	
CLO 2	Evaluate the suitability of message integrity, message authentication, and key management methods when applying in a given security scenario.	✓
CLO 3	Explain the setup, algorithms, and security issues in existing symmetric-key and asymmetric-key cryptosystems.	✓
CLO 4	Apply cryptography-based network security technologies in the design of networked information systems.	

Part 1 covers CLO 3 [70 points]**Question 1a [10 points]:**

Find the value of x for the following sets of congruence using the Chinese remainder theorem.

$$\begin{aligned}x &\equiv 2 \pmod{5} \\x &\equiv 3 \pmod{7} \\x &\equiv 10 \pmod{11}\end{aligned}$$

Question 1b [10 points]:

Find the result of the following using the square-and-multiply method.

$$3^{45} \pmod{77}$$

Question 1c [10 points]:

Determine whether the following integer passes the Miller-Rabin primality test:

$$561$$

Use base 2.

Question 2a [10 points]

In RSA cryptosystem, suppose that Shamir chooses $p = 13$ and $q = 11$, and calculates $n = 143$ and chooses $e = 7$ as his public key.

- i. Calculate the value of $\phi(n)$ and find out what will be the value of private key $d = ?$.
After this step, the key generation for Shamir will be complete.

Now imagine that Rivest wants to send a plaintext message with the integer value of $P = 83$ to Shamir. He uses Shamir's public key to encrypt the message.

- ii. Calculate the ciphertext C sent by Rivest to Shamir. Show all the steps.
- iii. Show all the steps which Shamir will apply to decrypt C in order to recover P .

Question 2b [10 points]

In ElGamal cryptosystem, given the prime $p = 31$, suppose Taher chooses $e_1 = 3$. Note that 3 is a primitive root in Z_{31}^* . Taher then chooses $d = 10$.

- i. Use the chosen e_1 and d to calculate e_2 for Taher.
- ii. Suppose Jamal chooses $r = 7$ to encrypt the message $P = 11$ to compute ciphertext (C_1, C_2) . Show all the steps for computing (C_1, C_2) .

Question 3 [20 points]:

Apply each of the following AES's transformations on the following plaintext assuming the use of same cipher key with all zeros.

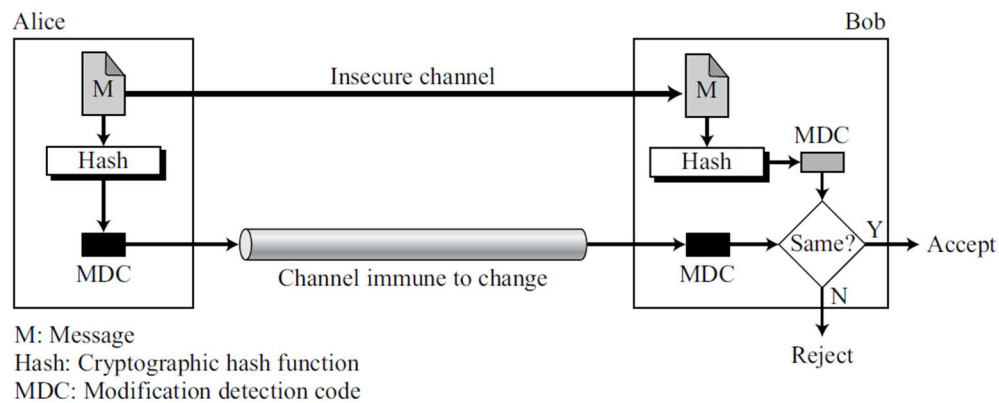
P: 80000000 80000000 80000000 80000000₁₆

- SubBytes
- ShiftRows

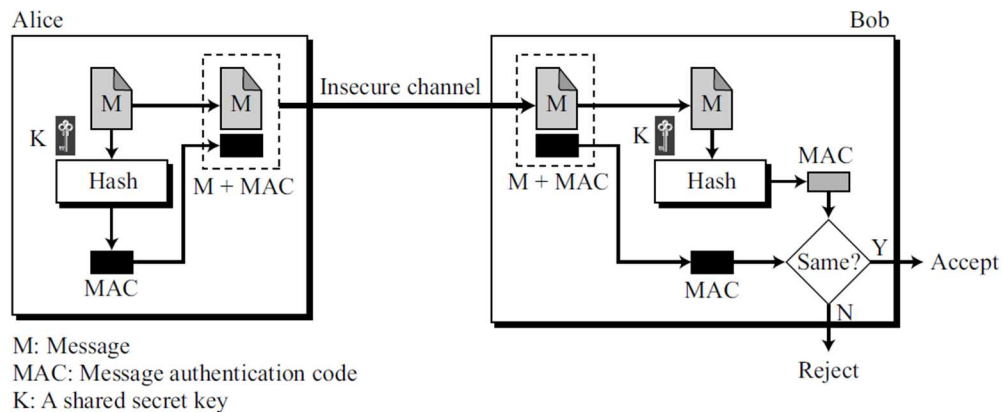
Note: Refer to the SubBytes transformation and other matrices on the last page.

Part 2 covers CLO 2 [30 points]**Question 4a [10 points]**

The following figure shows the process of checking message integrity with the help of message digests. Write an algorithm in pseudocode notation for the given diagram.

**Question 4b [10 points]**

The following figure shows the process of checking message integrity and message authentication with the help of message digests and a shared symmetric key K . Write an algorithm in pseudocode notation for the given diagram.

**Question 4c [10 points]**

Suppose Eve has intercepted the message M and the digest $h(K|M)$ of part 4b. If the size of the key allows exhaustive search, Eve may prepend all possible keys at the beginning of the message and make a digest of the $(K|M)$ to find the digest equal to the one intercepted. She

then knows the key and can successfully replace the message with a forged message of her choosing. This type of attack can deem MAC scheme useless for authentication.

- i. Discuss how can this type of attack on MAC can be avoided to make it useful for authentication?
- ii. Give one example of a specific MAC scheme which avoids this type of attack on key K.

Table 1. SubBytes transformation table

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	CB	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

